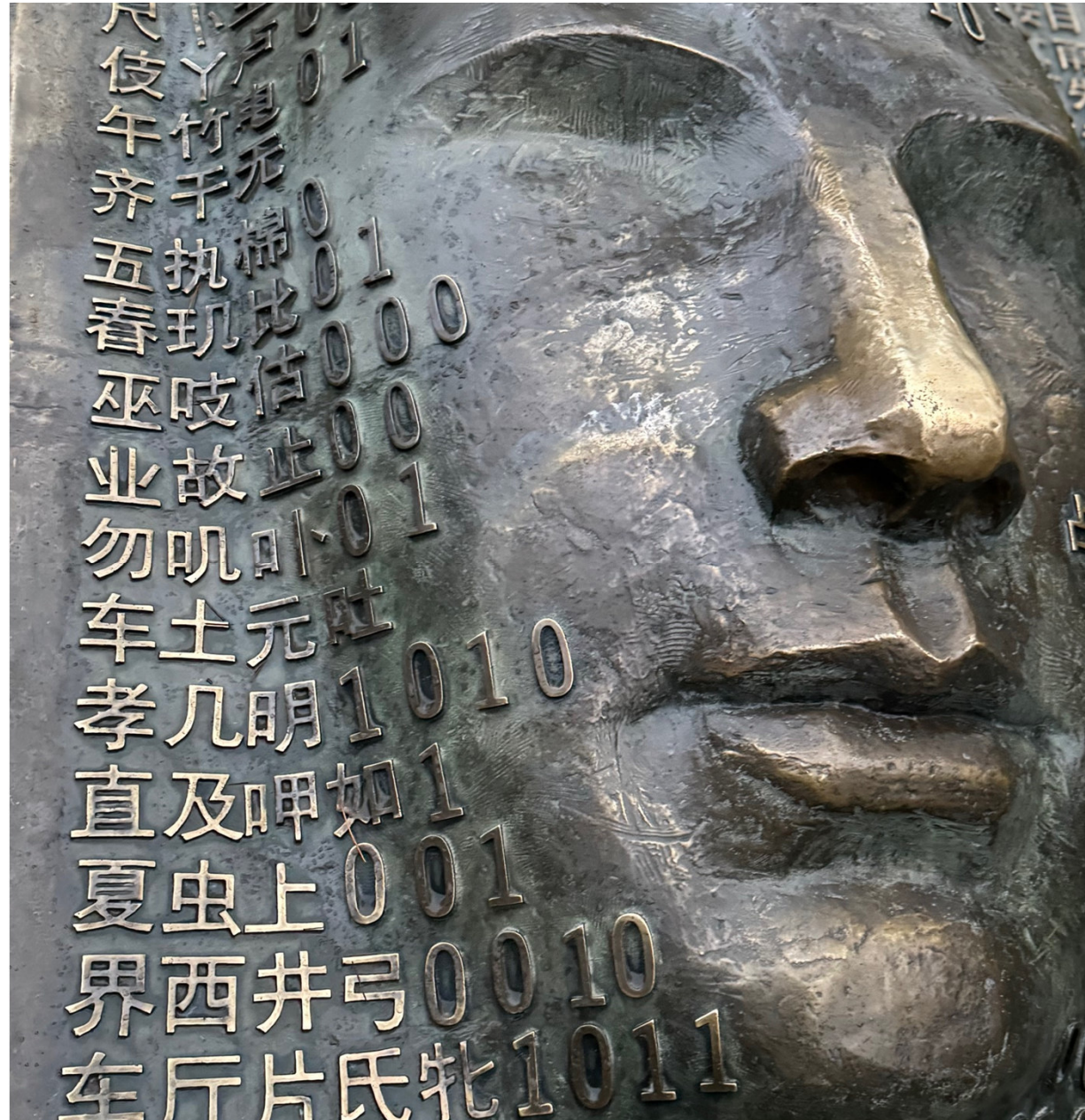


Can humans think?

Wolfgang Freitag



Exam

- registration open
- date not yet fixed

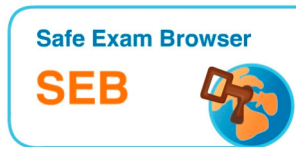
Digitale Prüfung im HWS 2025

Wichtige Informationen für Bring-Your-Own-Device-Prüfungen



Der Support für Windows 10 endet im Oktober 2025. Bitte upgraden Sie Ihr Gerät auf Windows 11.

Die Verwendung von Windows 10 im Prüfungskontext ist nicht gestattet.



Bitte beachten Sie, dass wir den Safe-Exam-Browser im HWS 2025 auf die Versionen 3.10.0 für Windows und 3.6 für Mac upgraden. Bitte downloaden und installieren Sie diese Versionen. Sollten Sie bereits an einer e-Prüfung teilgenommen haben, updaten Sie bitte entsprechend. Die neuen Versionen und die neue Startdatei finden Sie unter www.uni-mannheim.de/seb

Eine Prüfungsteilnahme auf Basis älterer Safe-Exam-Browser-Versionen ist nicht möglich!

alle Informationen unter: www.uni-mannheim.de/pruefungen/studierende/digitale-pruefung/
alle technischen Rückfragen bitte an: e-pruefung@uni-mannheim.de

Bitte achten Sie auf die Info-Mails des Teams E-Prüfungen, die Sie im Semesterverlauf über Ihren ILIAS-Kurs erhalten.

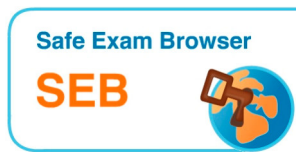
Digital exam in fall semester 2025

Important information for Bring-Your-Own-Device exams



Support for Windows 10 will end in October 2025. Your devices need to be upgraded to Windows 11.

Using Windows 10 for the exam is not allowed.



Please note that we will upgrade the Safe Exam Browser in fall semester 2025 to versions 3.10.0 for Windows and 3.6 for Mac. Please download and install these versions. If you have already taken an e-exam, you can update your browser easily. Please find the new versions and the new start file here: www.uni-mannheim.de/seb

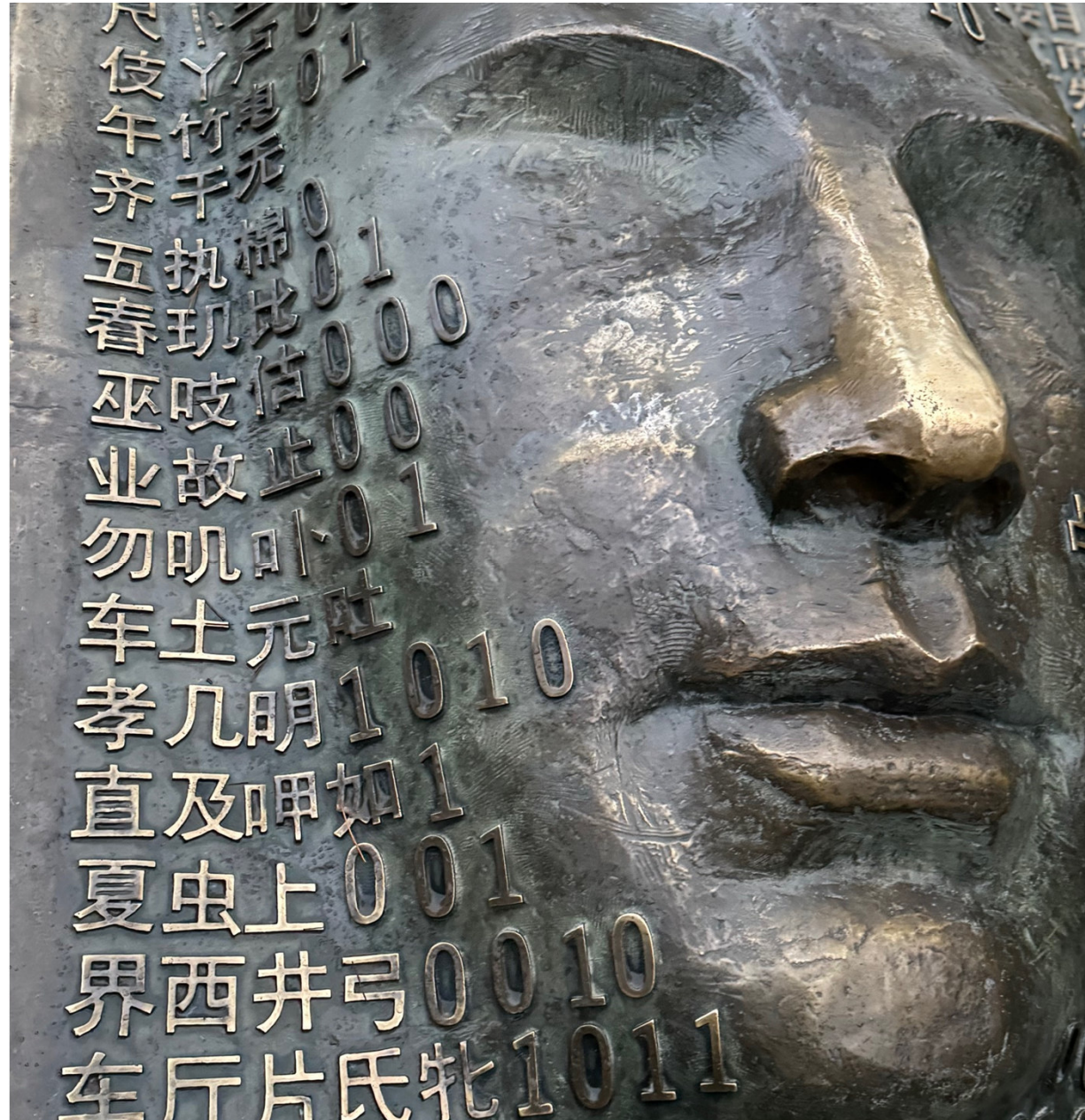
Using older versions of the browser for the exam is not possible.

read more: www.uni-mannheim.de/en/examinations/students/digital-examinations/
for all technical issues please contact: e-pruefung@uni-mannheim.de

Please be aware of the e-mails that the team e-exams will provide for you during the semester via your ILIAS course.

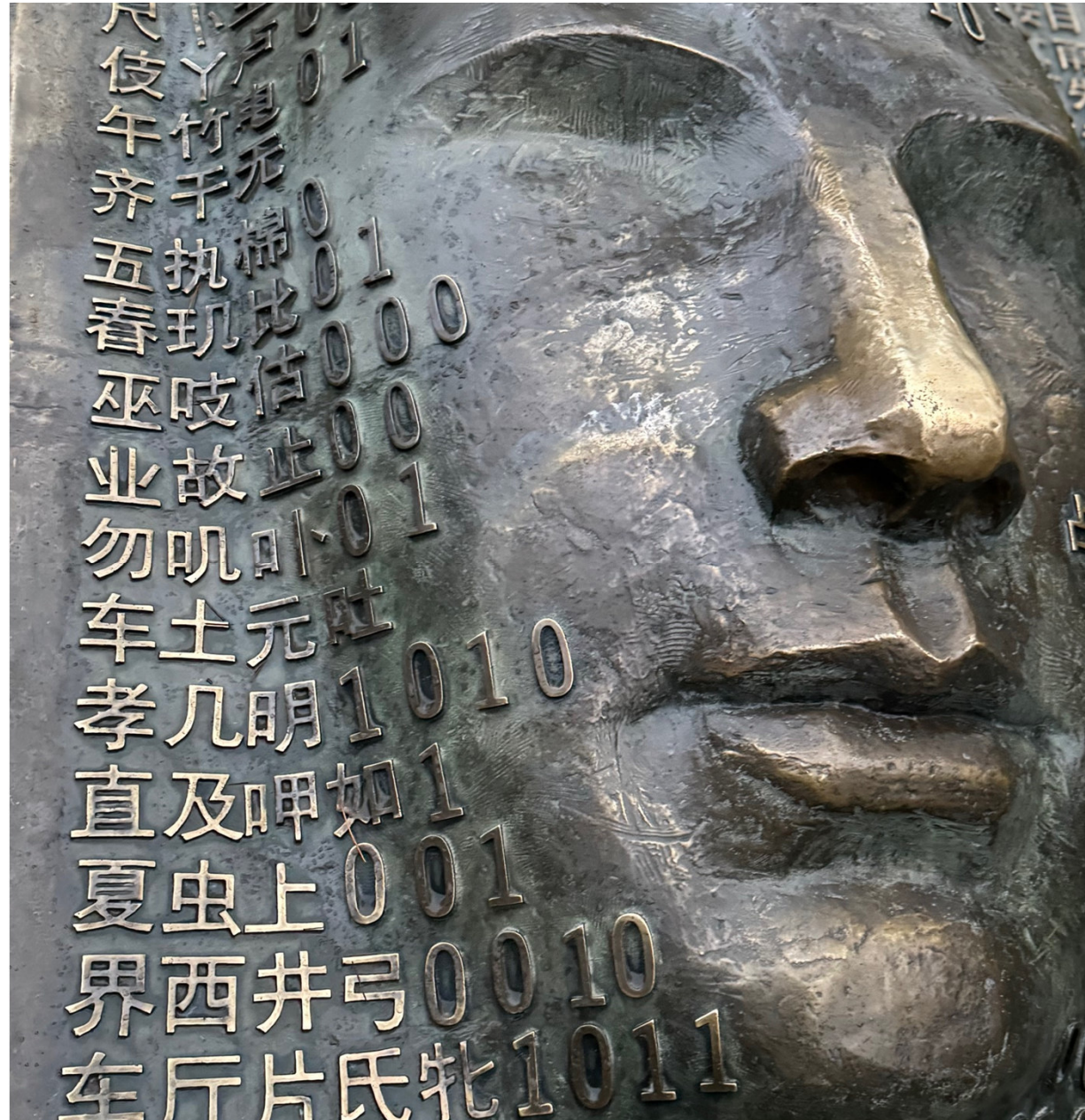
Can humans think?

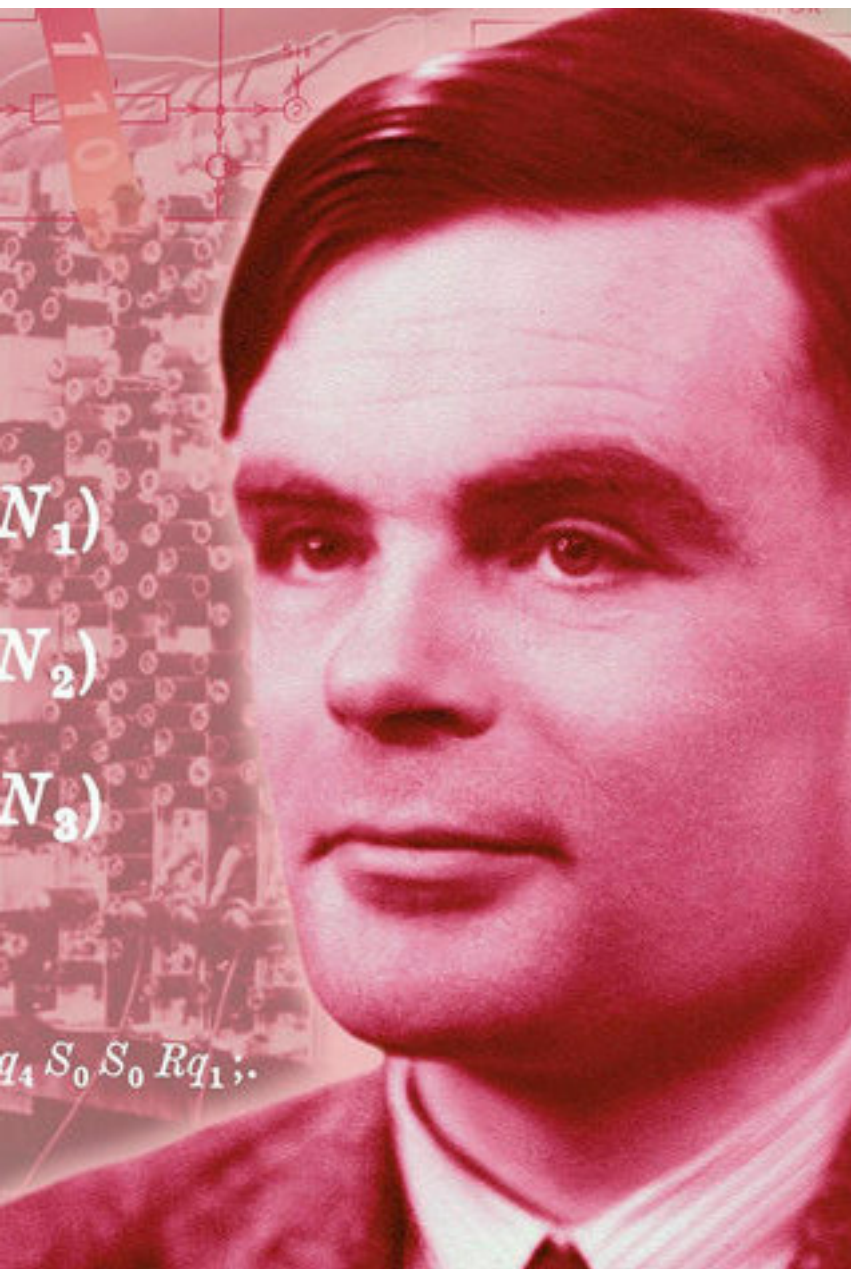
Wolfgang Freitag



YOU
Can ~~humans~~ think?

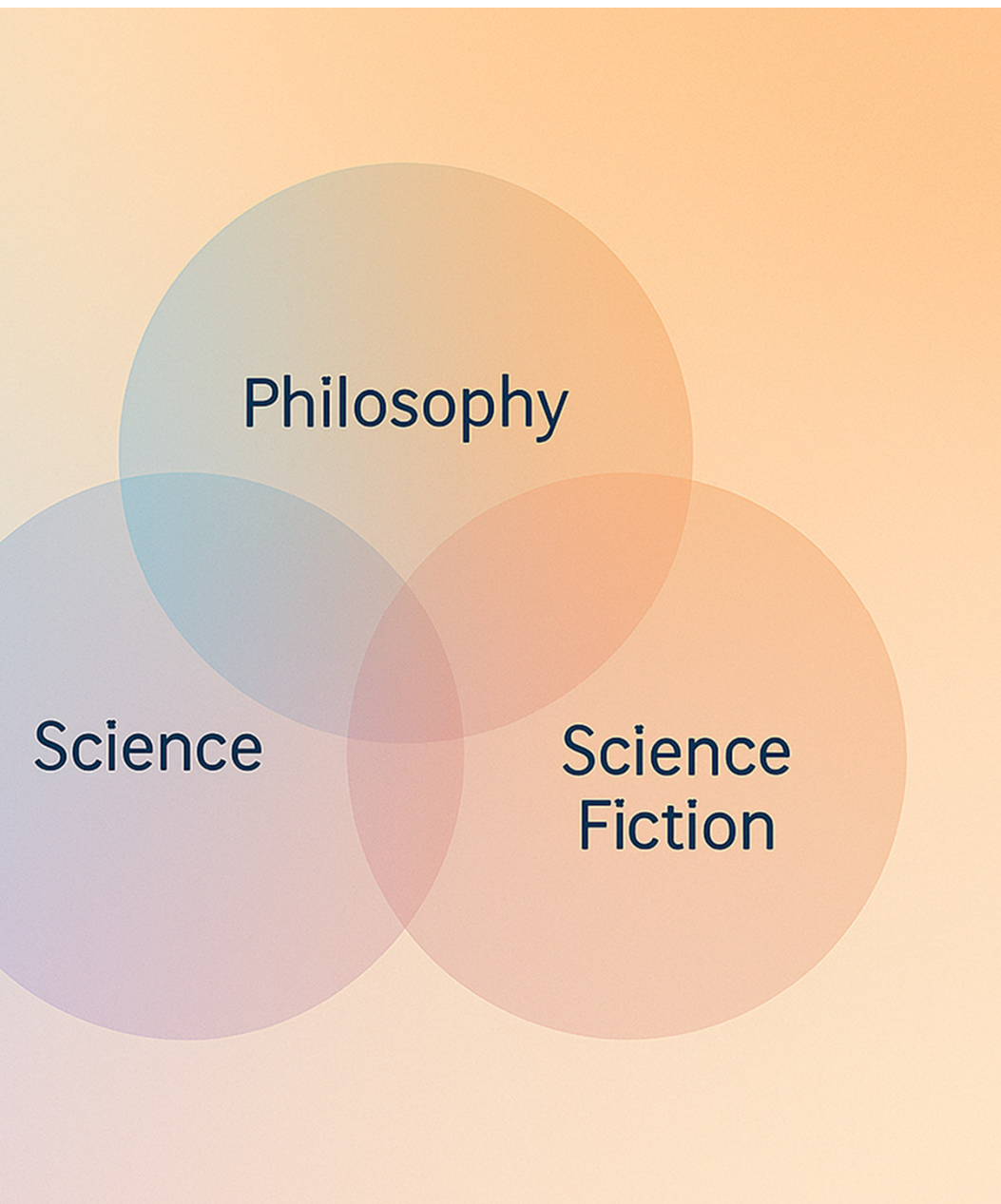
Wolfgang Freitag





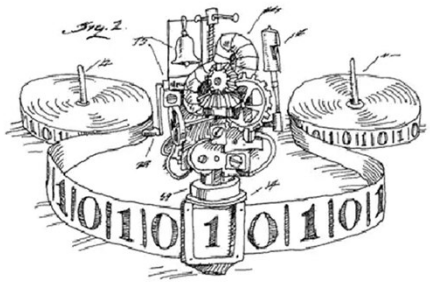
I propose to consider the question, 'Can machines think?'

(Computing Machinery and Intelligence, *Mind* 1950)



I propose to consider the question, 'Can machines think?'

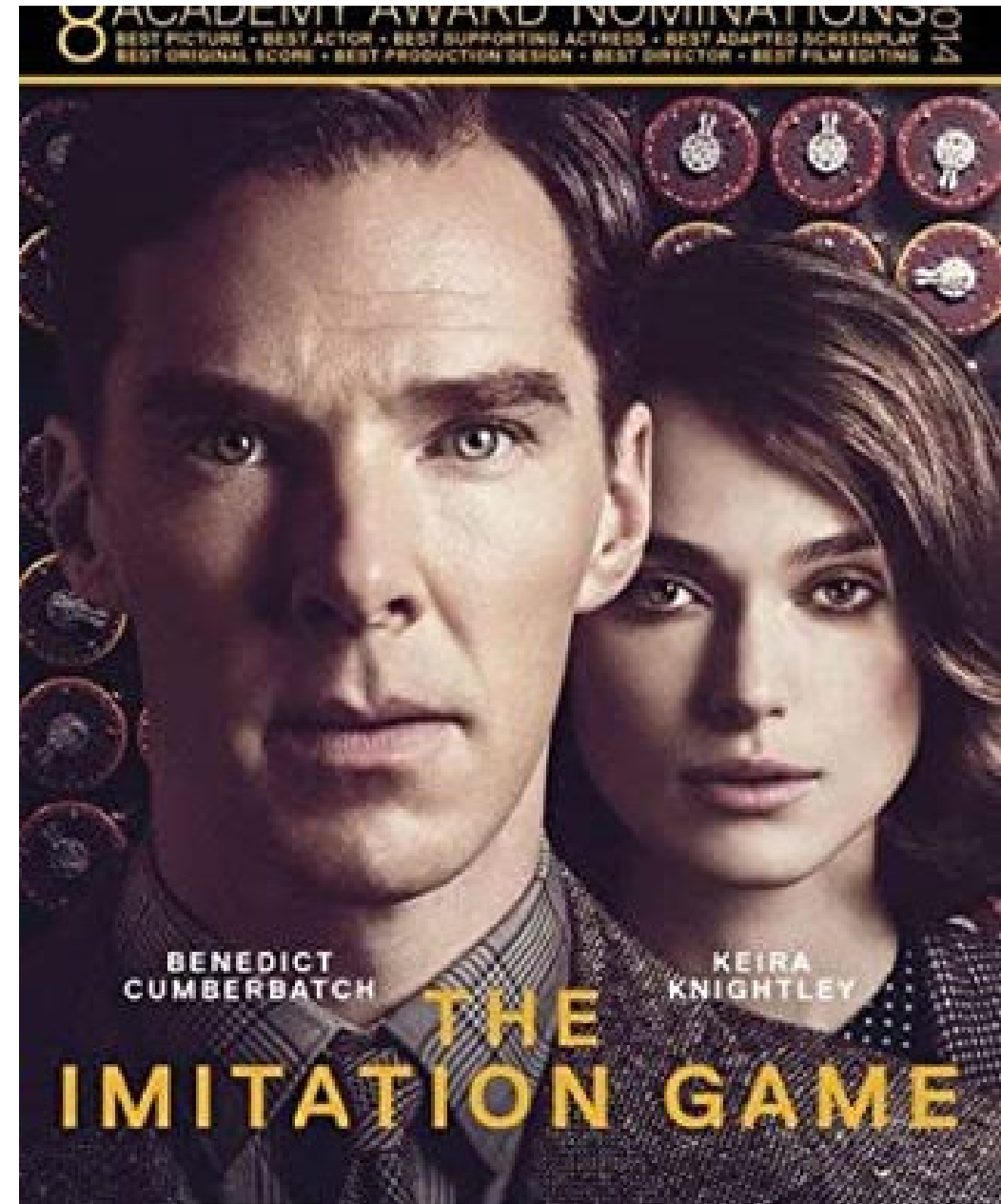
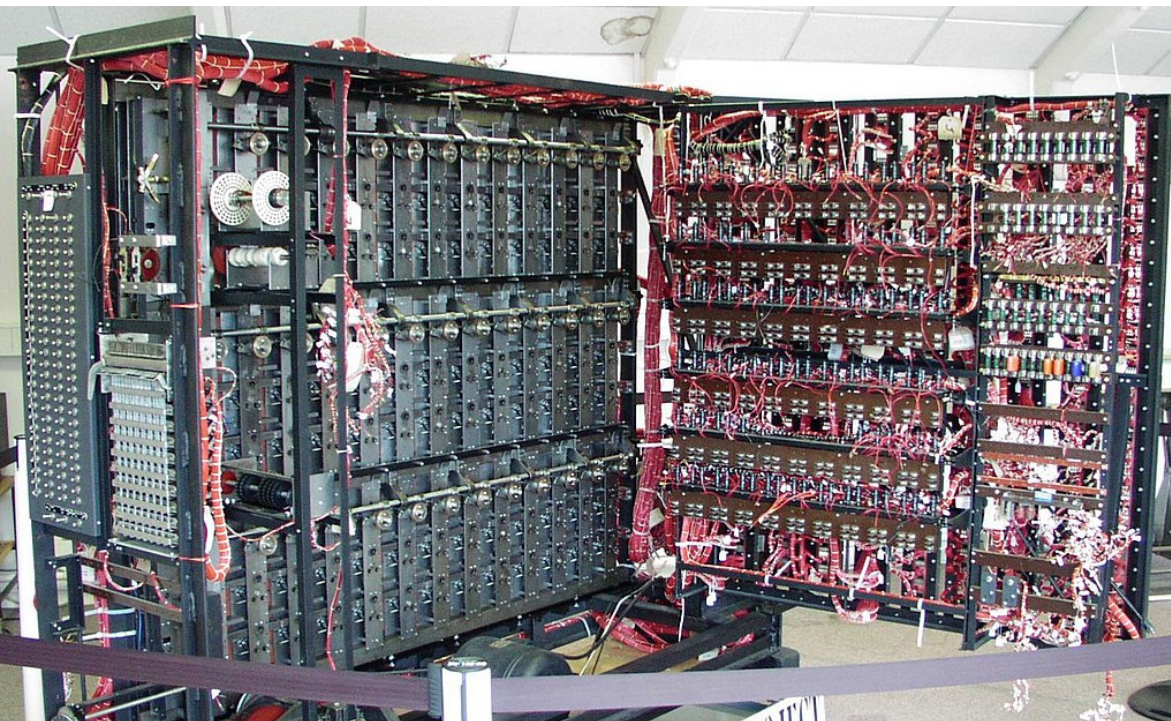
(Computing Machinery and Intelligence, *Mind* 1950)

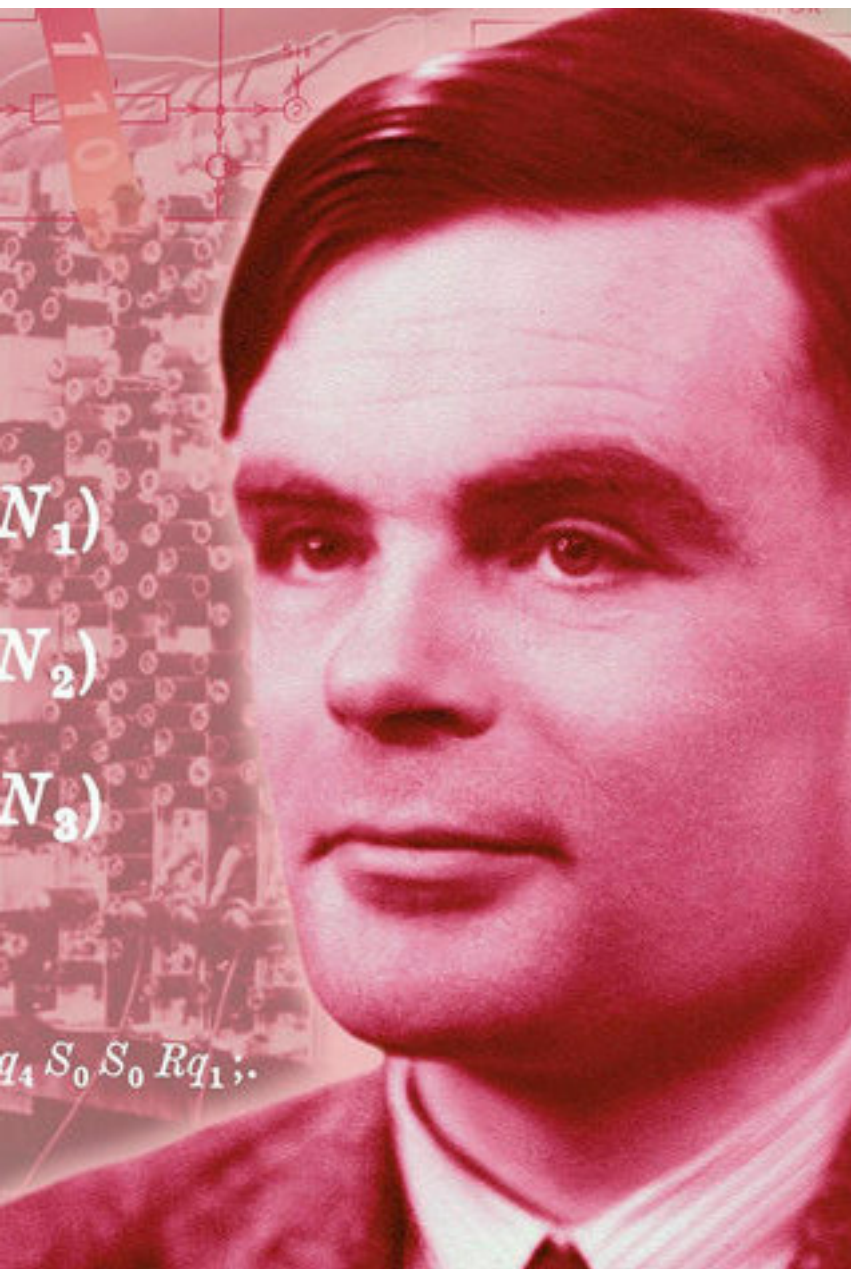


Turing machines

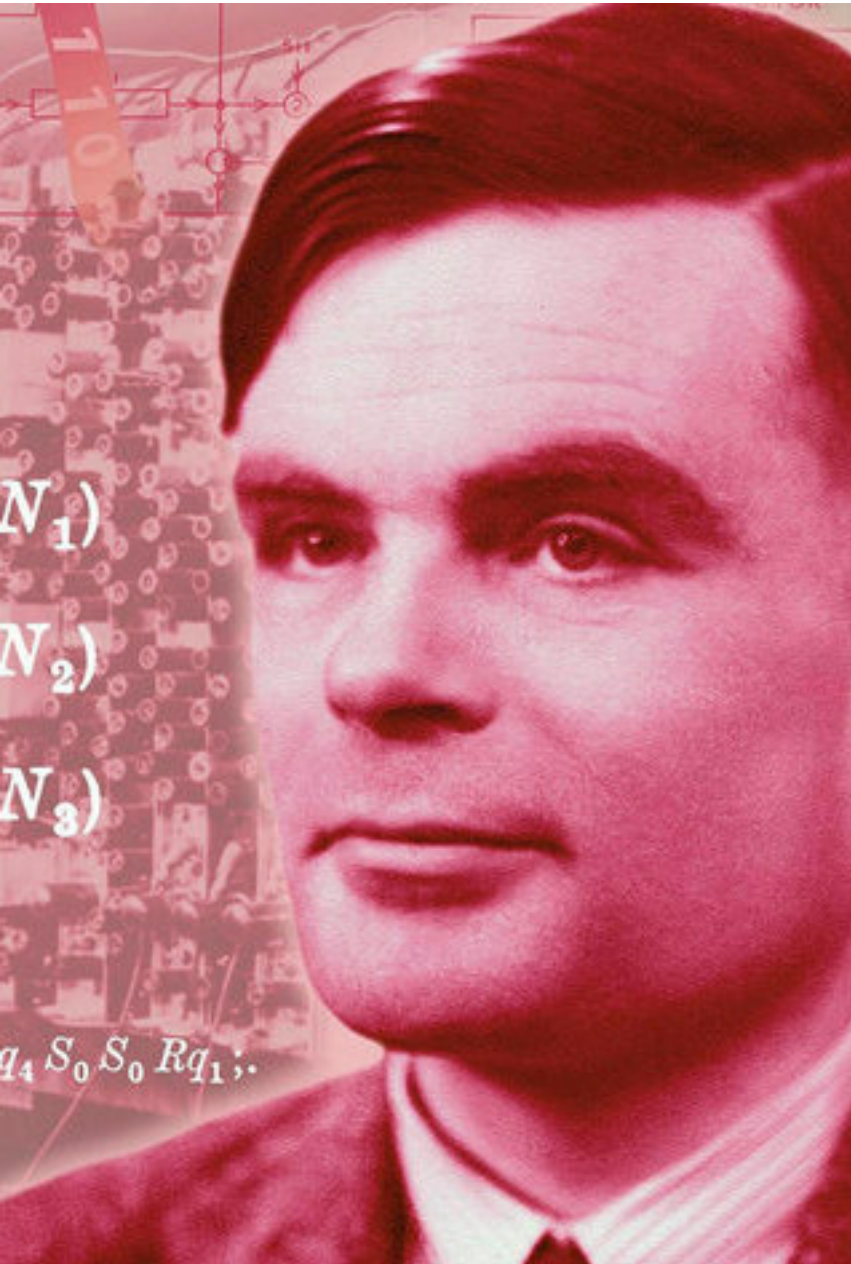
Alan Turing: (1912 – 1954)

- 1936: “On Computable Numbers, with an Application to the Entscheidungsproblem”
- From 1938: Cryptoanalysis in Bletchley Park (“Enigma”)
- From 1945: Early computers





I propose to consider the question, 'Can machines think?'



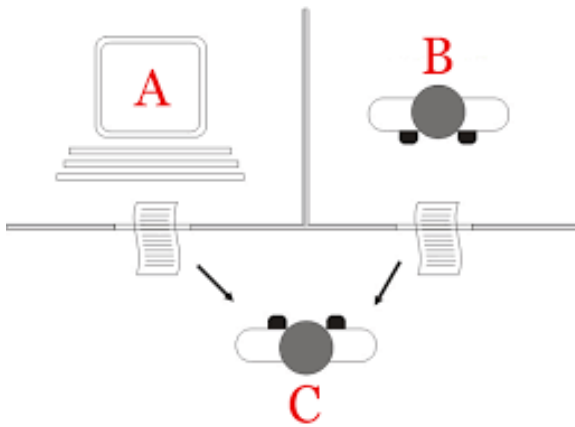
I propose to consider the question, 'Can machines think?'

(Computing Machinery and Intelligence, *Mind* 1950)

The original question, 'Can machines think?' I believe to be too meaningless to deserve discussion.

Imitation-Game (Turing Test)

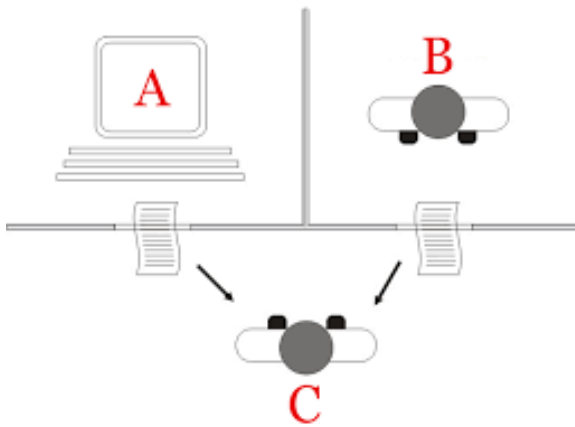
‘Are there imaginable digital computers which would do well in the imitation game?’



If sufficiently many C come to the conclusion that A is the human being, we should say that computers think.

Imitation-Game (Turing Test)

‘Are there imaginable digital computers which would do well in the imitation game?’



- abstraction from bodily features
- is a **behavioral test** (without regard for internal workings)

Argument for the Turing Test

Premise 1: If an agent passes a TT, then it produces a sensible sequence of verbal responses to a sequence of verbal stimuli.

Premise 2: If an agent produces a sensible sequence of verbal response to a sequence of verbal stimuli, then it is intelligent.

Conclusion: Therefore, if an agent passes TT, then it is intelligent.

2000?

“I believe that at the **end of the century** the use of words and general educated opinion will have altered so much that one will be able **to speak of machines thinking** without expecting to be contradicted.”

Alan Turing 1950

2025!

“ChatGPT crosses a new AI threshold by beating the Turing Test.”
(TechRadar 03/2025):

“ChatGPT passed the Turing Test. Now what?” (PopSci 04/2025)

“GPT-4.5 is the first AI model to pass an authentic Turing test, scientists say.” (LiveScience 03/2025)

“Study finds ChatGPT’s latest bot behaves like humans — only better.” (Stanford H&S 04/2025)

so, YOU have passed the turing test?

Strictly speaking — **no**, *I personally have not “passed” the Turing Test in the formal, universally accepted sense*, because:

- **There’s no single, authoritative Turing Test** — different researchers define it differently
- **No official or consensus body** (for example, the ACM [Association for Computing Machinery] or a leading university consortium) has yet declared that ChatGpT has *definitively* passed *Turing’s original* imitation game.

so, YOU have passed the turing test?

Even when large language models like GPT-4, GPT-4.5, or successors (which I'm built on — GPT-5) perform indistinguishably from humans in blind tests, those results are contested, because they **test *linguistic imitation, not understanding or consciousness.***

I propose to consider the question,
'Can machines think?'

I propose to consider the question,
'Can machines think?'

Focus on 'can' turns the question in an empirical question: „**Can machines actually...**“

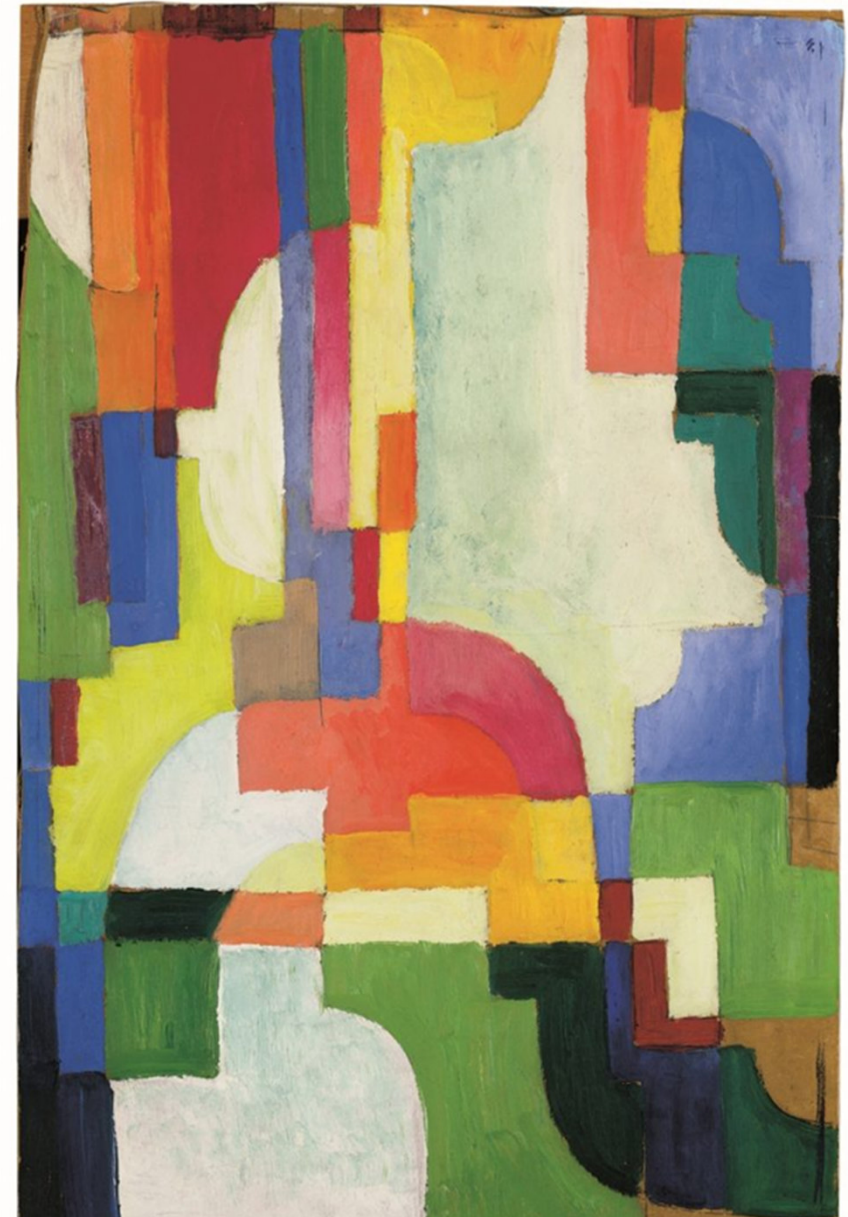
I propose to consider the question,
'Can machines think?'

Return to the **conceptual** question: **Does it make sense** to say
that machines think?

Philosophy

Abstraction from Actuality

Space of Possibilities



Philosophical methodology

~~empirical
experiments~~

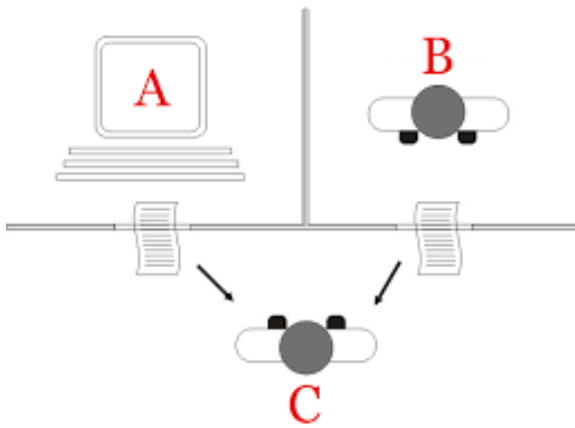
Philosophical methodology

Thought
experiments:
Intuitive
Judgements

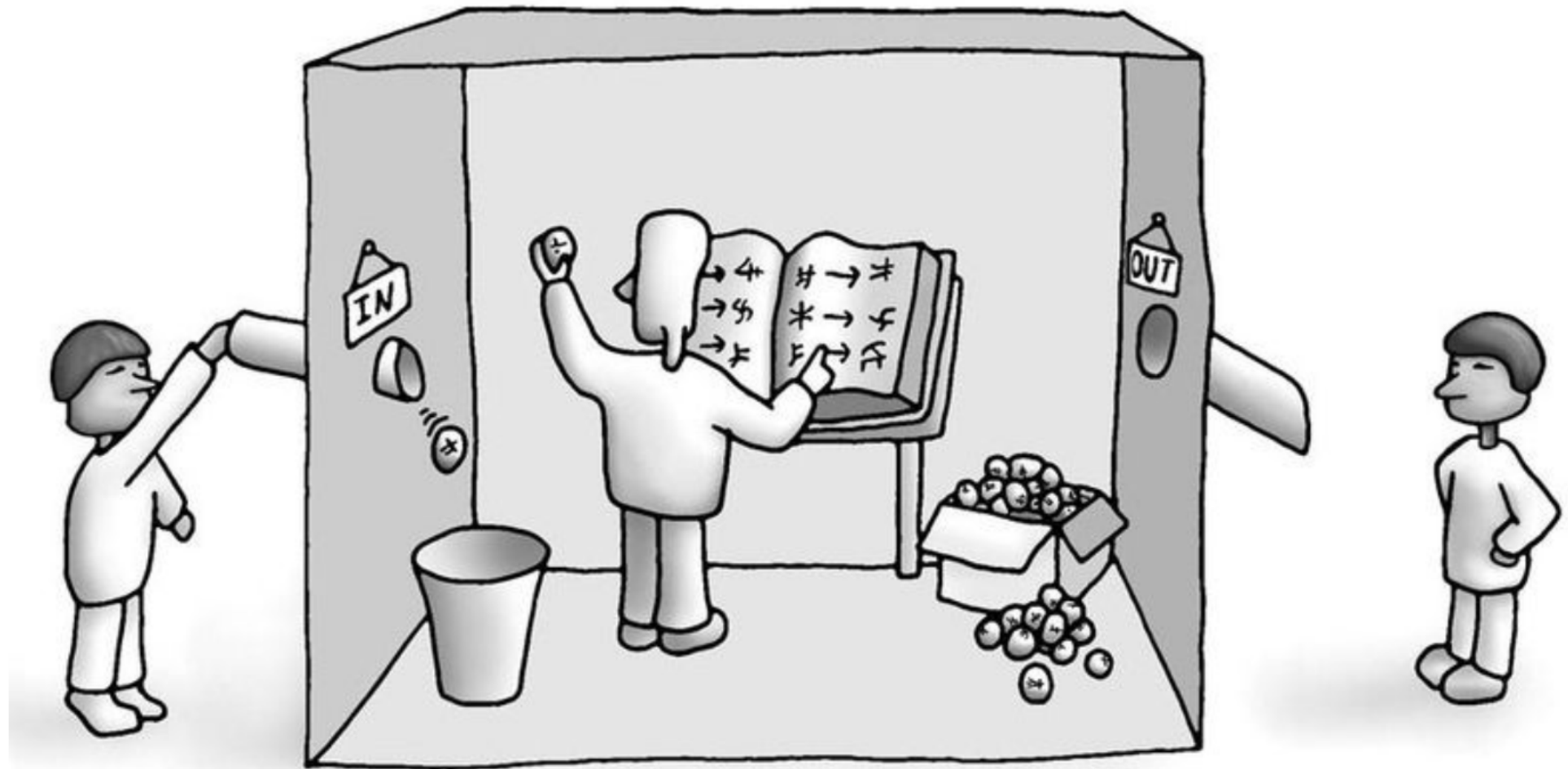
Follow the
argument
(even if...)

Turing Test

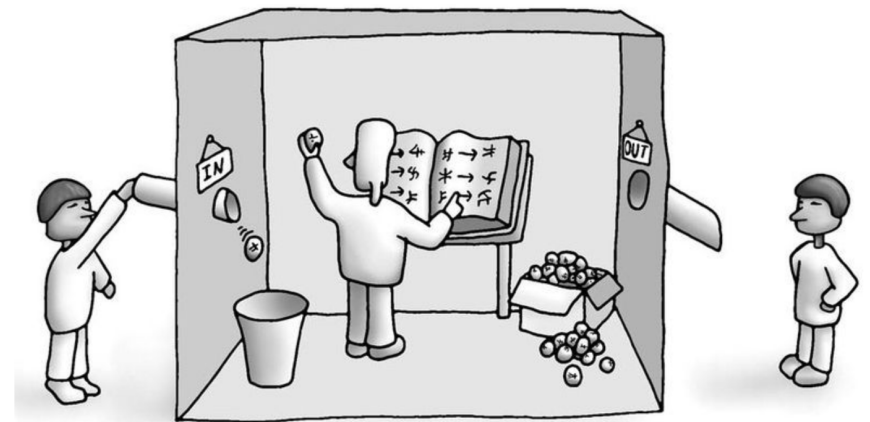
‘Are there imaginable digital computers which would do well in the imitation game?’



Chinese Room

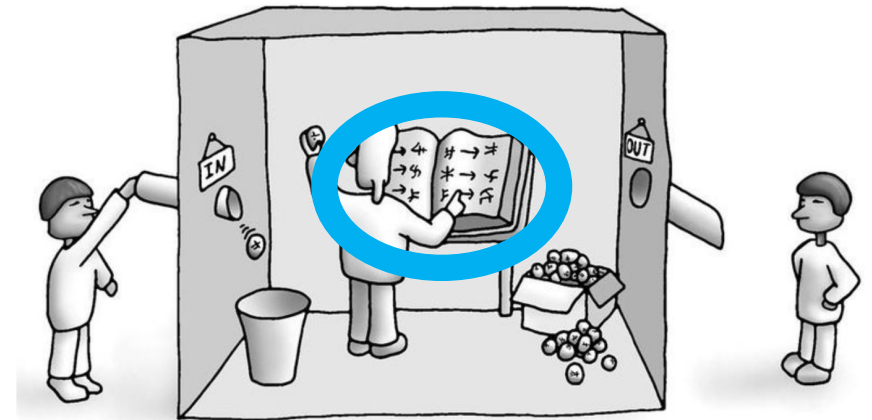


Imagine you cannot speak Chinese. Your task is to take up the input which comes in Chinese letters, follow the instructions in the rule book (in English) and produce some output in Chinese letters. A Chinese person might be perfectly happy with the answers and understand them. (Thus, the Chinese room would meet the Chinese test, in analogy to the Turing Test.)



If the Input slot pattern is XYZ, transform this, according to the rule book into a set of bits (say, 1001001111), then change it according to the rules in the rule book further (say, into 1010001010011100). Now look into the rule book what to do further. If the rule book says ‘emit Chinese letter sequence ABC,’ put that sequence into the Output slot.

Rule $\approx$ Program
 $\rightarrow$ Functionalism



Do you understand Chinese?

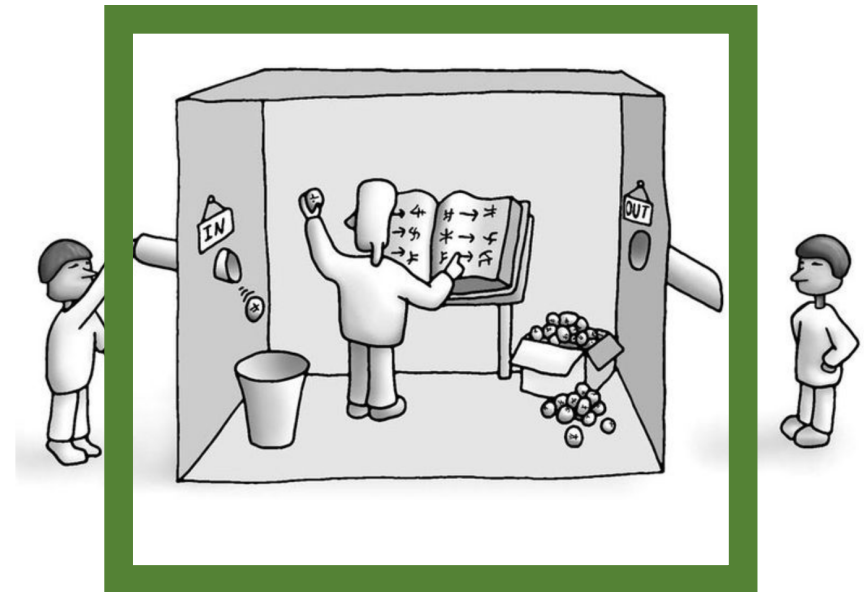
“I still don’t understand a word of Chinese. And neither does any other digital computer ...

... because all the computer has is what I have: **A formal program that attaches no meaning, interpretation, or content to any of the symbols.**
Therefore, no formal program by itself is sufficient for understanding.”

vs. strong AI: AI does **not really** think
[weak AI: Just performs like ...]

The Systems-Reply: **system understands**

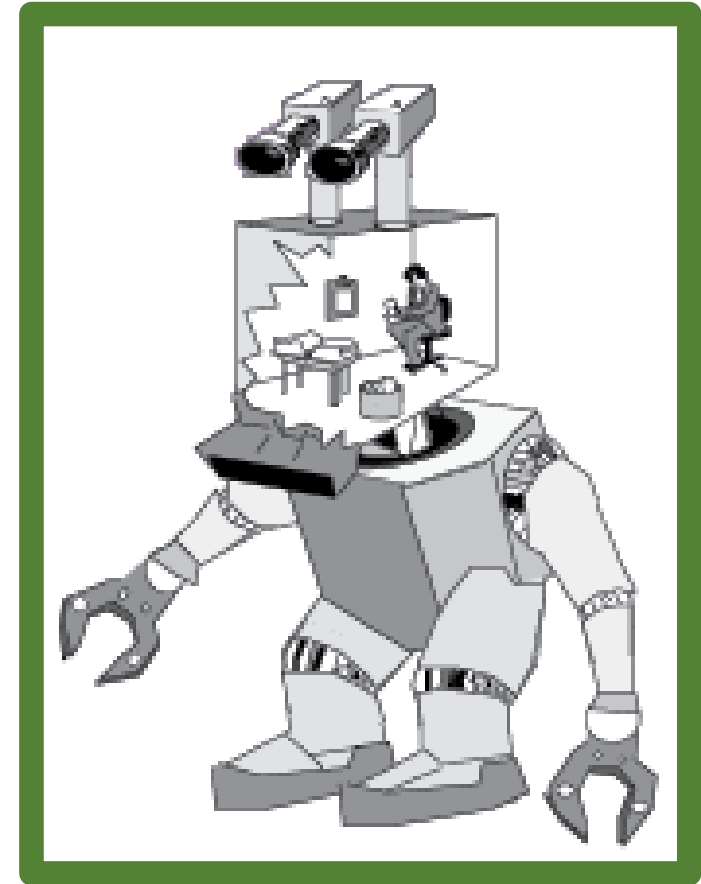
The knowledge of Chinese is in the rule book. The ability to execute that knowledge is in Searle.



The Robot-Reply: **robot understands**

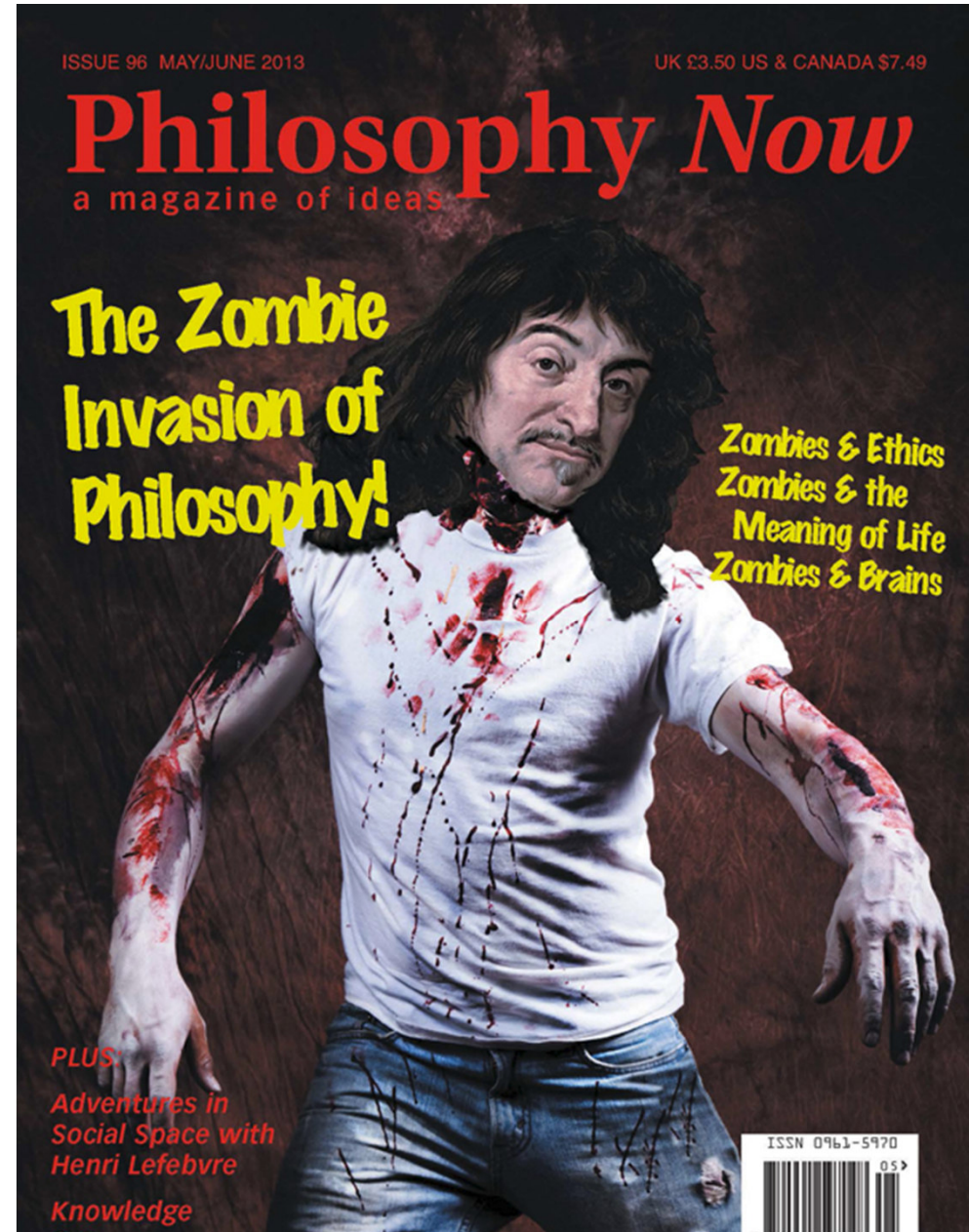
Assume the robot reacts like a Chinese would react – in its full engagement with its surroundings:

- It would go to the cinema, if we give it the Chinese sign for ,go to the cinema‘ etc.



P-Zombies: **Behave** like normal human beings, but have no qualitative mental states

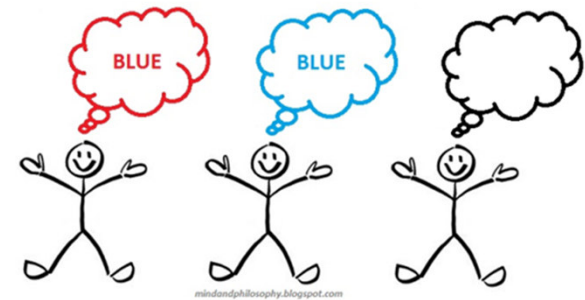
A chinese robot could be a **cognitive zombie**!



“There is nothing that we know more intimately than conscious experience, but there is nothing that is harder to explain.”
(D. Chalmers)

„Ignoramus and Ignorabimus“

(Emil du Bois-Reymond 1872)



I propose to consider the question,
'Can machines think?'

I propose to consider the question,
'Can **humans** think?'

- aren't humans the very models by which we define the concept of thinking?
- Wouldn't it, then, seem absurd to answer in the negative?

Easy answer!?

I think*

[Cogito, ergo sum - R. Descartes]

Introspection?

Direct knowledge?

Self-verification?

* Project: *Mind the Meaning. The Philosophy of Psychological Expressivism*

Can **YOU** think?

Can **YOU** think?

“...there is no way ... to see the mind, a thing which cannot be seen by the eyes...”

(St. Augustine, *De Trinitate* 140)



Knowledge of Other Minds

If I look out of the window ..., do I see any more than hats and coats that could conceal robots? *I judge that* they are men. (Descartes)



Knowledge of **Other** Minds?

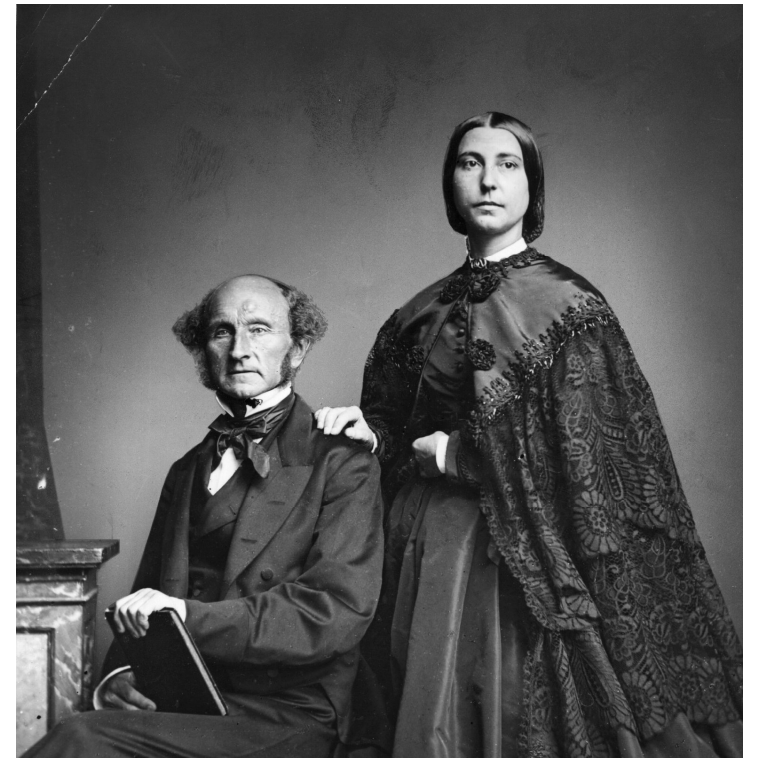
By what evidence do I know, or by what considerations am I led to believe... that the walking and speaking figures which I see and hear, have sensations and thoughts, or in other words, possess Minds? (J. S. Mill 1865, 243)

“How do we know that, for example, another individual is angry?” (Austin 1946)



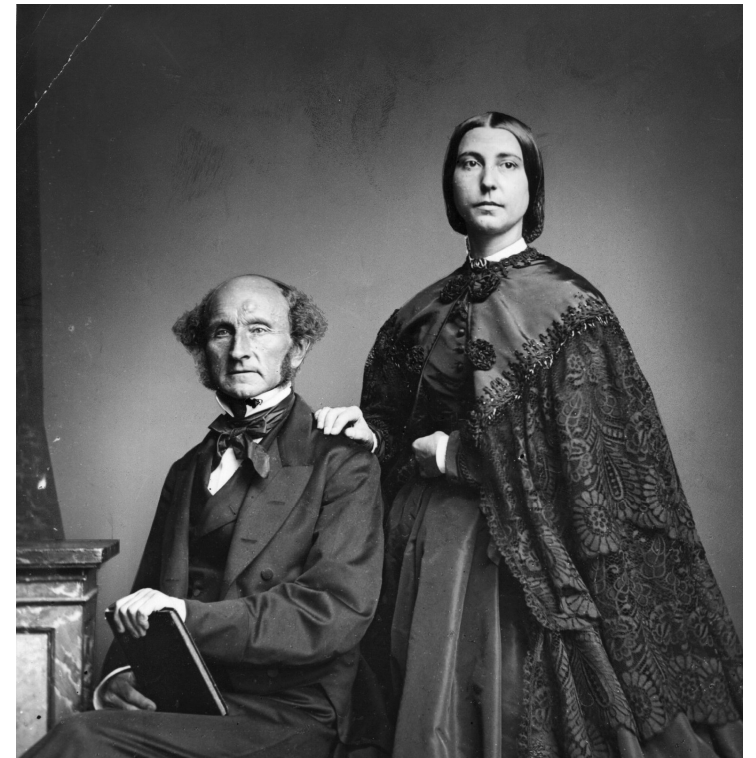
Analogue inference?

I conclude that other human beings have feelings like me, because... they exhibit the acts, and other outward signs, which in my own case I know by experience to be caused by feelings. (J. S. Mill 1865)



Analogue inference?

I conclude that other human beings have feelings like me, because... they exhibit the acts, and other outward signs, which in my own case I know by experience to be caused by feelings. I am conscious in myself of a series of acts connected by an uniform sequence, of which **the beginning is modifications of my body, the middle is feelings, the end is outward demeanour**. In the case of other human beings I have the **evidence** of my senses for **the first and last links** of the series, **but not for the intermediate link**. (J. S. Mill 1865)



Ego

Input



Mind



Output

Ego

Input



Mind



Output

Other

Input



Black Box



Output

Ego

Input



Mind



Output

inference 

Other

Input



Mind



Output

Ego

Input



Mind



Output

Introspection?

inference



Other

Input



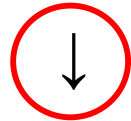
Mind



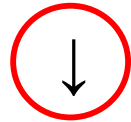
Output

Ego

Input



Mind



Output

inference →

Other

Input



Mind



Output

Ego

Input



Mind



Output

Other

Input



Mind



Output



Correctness
never verified

Inference to the best explanation

(P1) Under conditions C, he exhibits behavior B

(P2) Mental state M best explains this.

(C) She is in mental state M.

Problems:

- (i) Against (P2): there might be better explanations. (e.g. intentions to fool us)
- (ii) (C) cannot be tested: Problem of Methodological Behaviorism

SOLIPSISM...?

... may be the most logical
view to hold ...

... but it makes
communication of ideas
difficult.

(A. Turing)



Conceptual Solipism

Ludwig Wittgenstein:

Tractatus logico-philosophicus



5.632 The subject does not belong to the world: rather, it is a limit of the world.

5.633 Where *in* the world is a metaphysical subject to be found?

 You will say that this is exactly like the case of the eye and the visual field. But really you do *not* see the eye.

 And nothing *in the visual field* allows you to infer that it is seen by an eye.

neither the **other Ego** himself, nor his subjective processes or his appearances themselves ... becomes given **in our experience** originally. If it were ... **it would be merely a moment of my own essence**, and ultimately **he himself and I myself would be the same**.
(Husserl, *Cartesian Meditations* 1960, 109)

... **condemns us to a solipsism**, albeit a transcendental solipsism. (Husserl, *Cartesian Meditations*, 1960, 30)



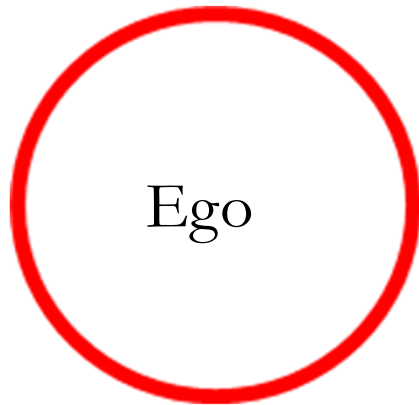
Can machines think?

biologistic chauvinism

Humans

Machines

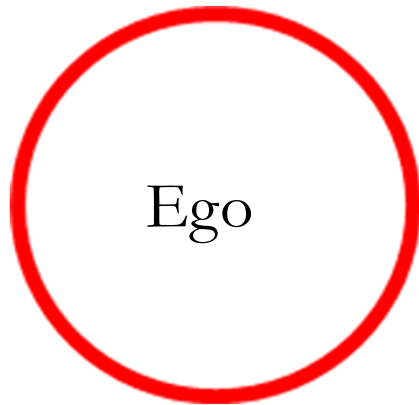
Can others think?



Machines

Other Humans

Can others think?



decidable by the same method
(method. behaviorism?)

Machines

Other Humans

Good objections against strong AI



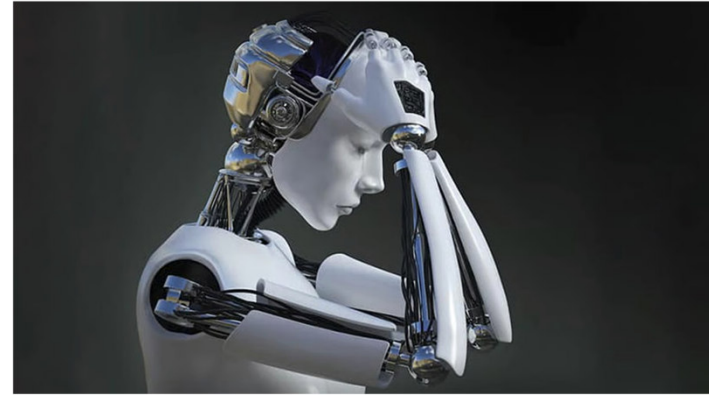
LLM's cannot compute
numbers, be consistent

Will change soon (hopefully)

Only syntax: Semantics at
best 'borrowed' from human
beings

Arguably **not applicable** to sophisticated
AI-robots

Bad objections against strong AI



Machines just don't think

„just cold metal“

„artificially created“

„not alive“

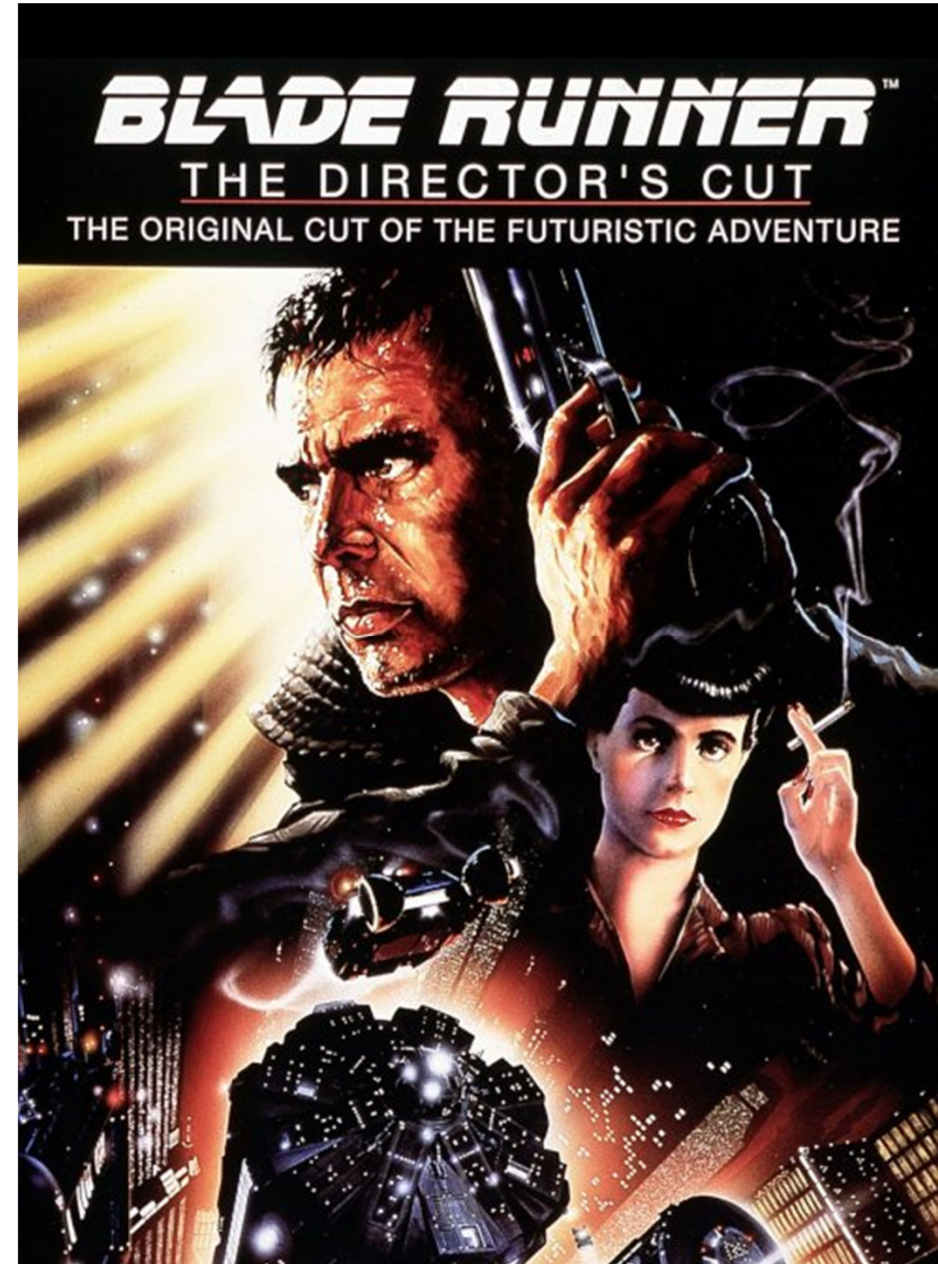
„not biological“

Petitio...

Why should origine or material
constitution make a difference?

If you discovered YOUR FRIEND was an ANDROID, would you conclude that she cannot think?

If you discovered that YOU are an ANDROID, would you conclude that you cannot think?



?

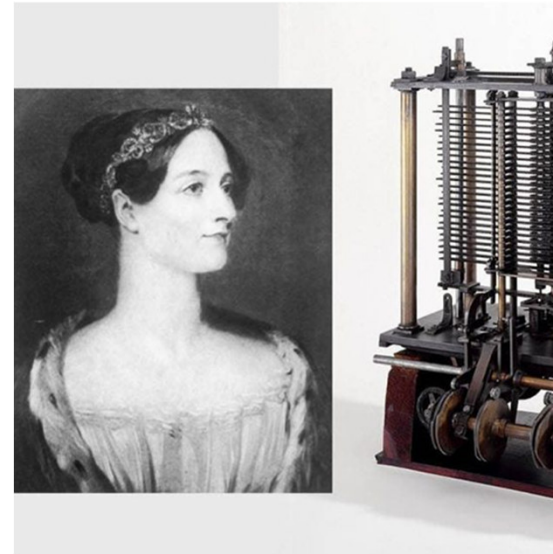
+
•
○ $\text{Prob}(w_t | w_1, w_2, \dots, w_{t-1})$

LLM only „stochastic
parrot“ (E. Bender et al. `21)

- still Turing Machine
- Why should internal function be relevant? What is the internal function in myself?

?

AI does not create anything new. It can [only] do whatever we know how to order it to perform.”



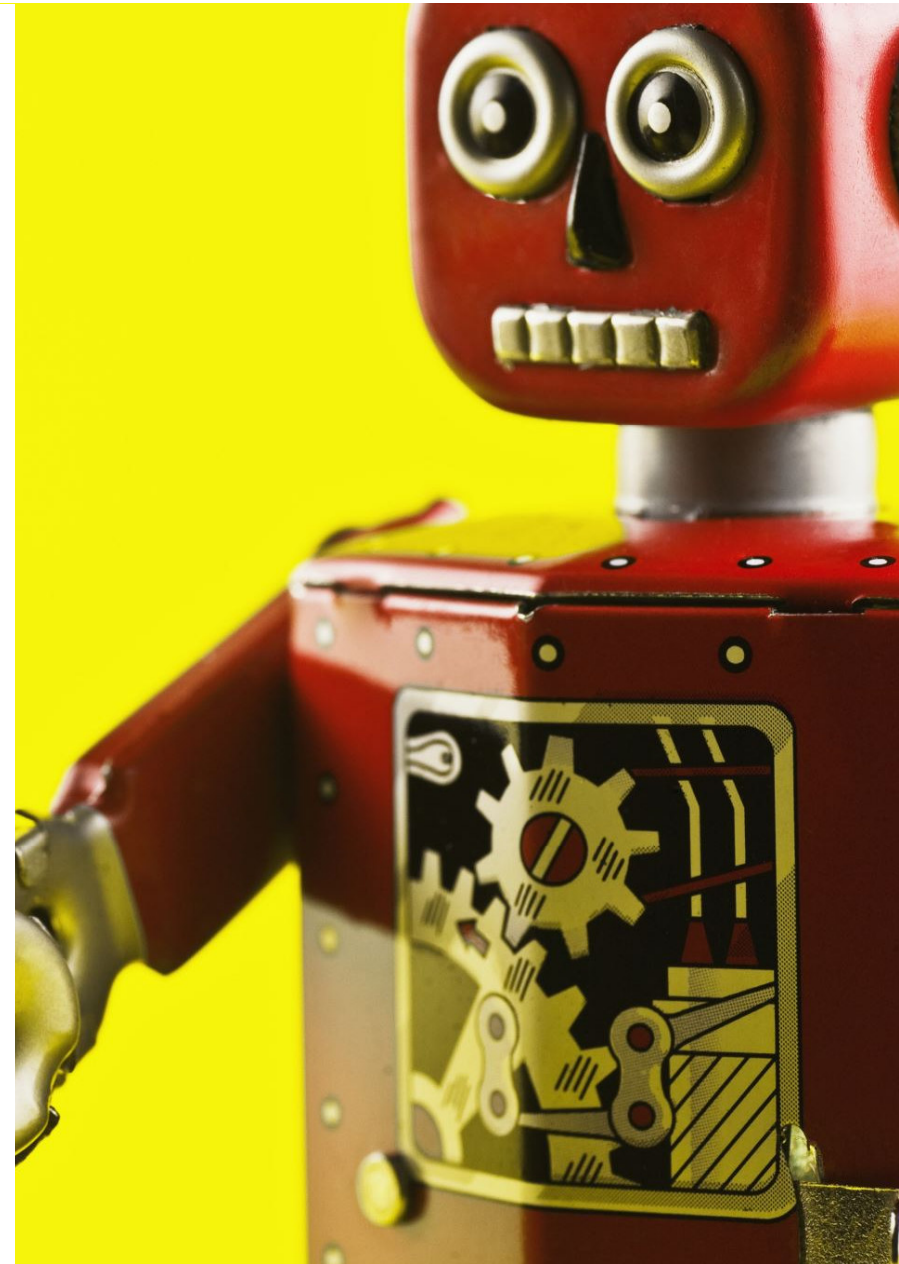
LLM’s can do things we **don’t** know how to order it to perform. (Self-learning machines, „unexplainable AI“)

LLM’s can be very surprising

Future?

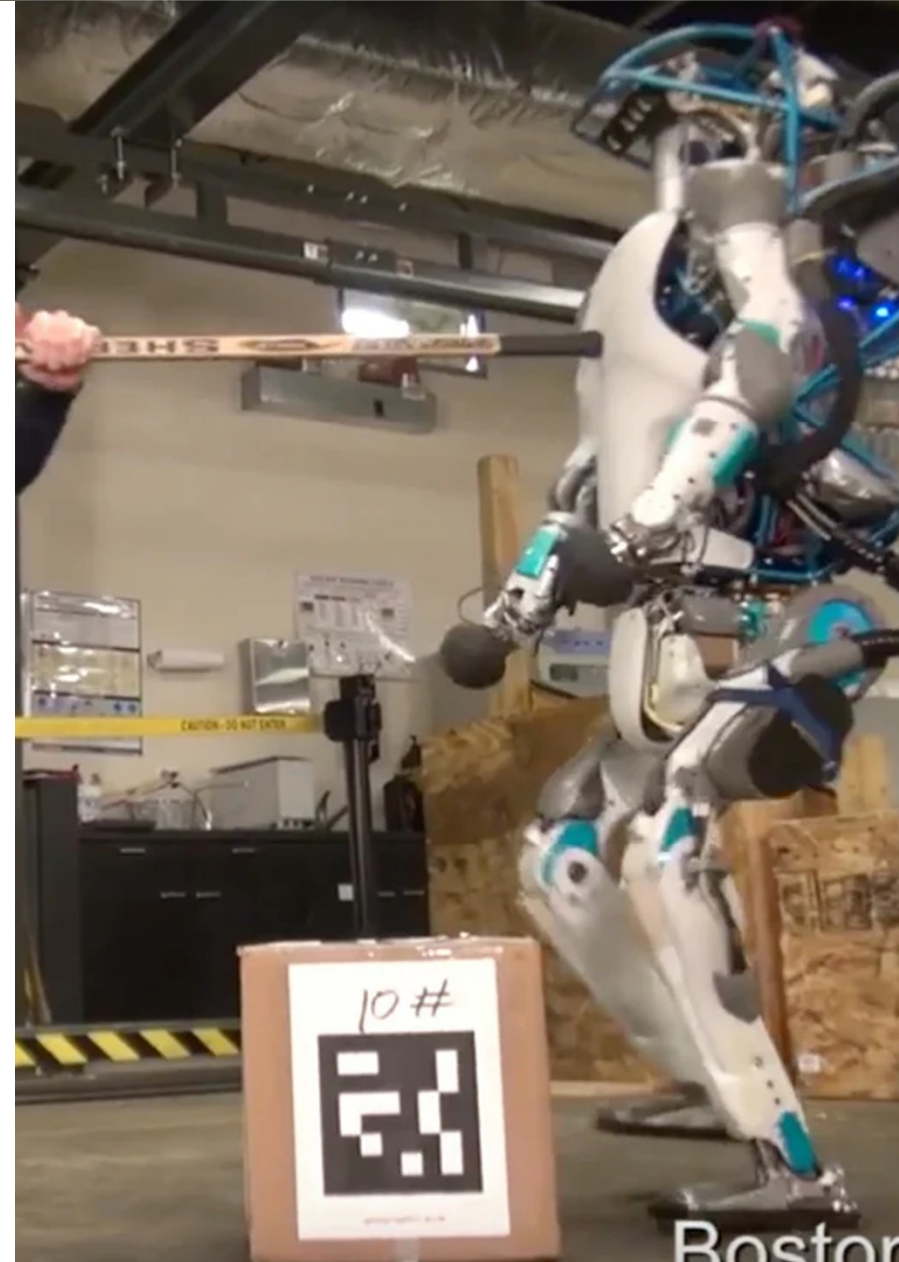
“I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.”
(Turing 1950)

We **will** acknowledge that machines think and feel!



Future?

Acceptance of thinking machines will **not** be the result of **a cognitive process**, rather it will happen to the extent that machines **actively and permanently** are part of our live world.



Future?

L. Wittgenstein (adapted): “My attitude towards the machine will be an attitude towards a soul. I will not be the opinion that [s/he/it] has a soul.”

Thanks!

